A7 Markov Decision Processes with Undiscounted Rewards

We focus on Markov Decision Processes (MDPs) with undiscounted rewards, since the relevance but also the problems of a delayed reward can be considerably decreased by discounting it. Using discounted rewards both the bias correction in TD as well as the variance of MC are greatly reduced. The correction amount decreases exponentially with the delay steps, and also the variance contribution to one state decreases exponentially with the delay of the reward.
MDPs with undiscounted rewards are either finite horizon or process absorbing states without reward. The former can always be described by the latter.

A7.1 Properties of the Bellman Operator in MDPs with Undiscounted Rewards

At each time t the environment is in some state $s = s_t \in \mathcal{S}$. The agent takes an action $a = a_t \in \mathcal{A}$ according to policy π, which causes a transition of the environment to state $s' = s_{t+1} \in \mathcal{S}$ and a reward $r = r_{t+1} \in \mathcal{R}$ for the agent with probability $p(s', r \mid s, a)$.
The Bellman operator maps a action-value function $q = q(s, a)$ to another action-value function. We do not require that q are Q-values and that r is the actual reward. We define the Bellman operator T^π for policy π as:

$$\mathrm{T}^\pi\left[q\right](s,a) \;=\; \sum_{s',r} p(s',r \mid s,a) \left[r \;+\; \sum_{a'} \pi(a' \mid s')\, q(s',a') \right] . \tag{A276}$$

We often rewrite the operator as

$$\mathrm{T}^\pi\left[q\right](s,a) \;=\; r(s,a) \;+\; \mathrm{E}_{s',a'}\left[q(s',a')\right] \;, \tag{A277}$$

where

$$r(s,a) \;=\; \sum_r r\, p(r \mid s,a) \;, \tag{A278}$$

$$\mathrm{E}_{s',a'}\left[q(s',a')\right] \;=\; \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\, q(s',a') \;. \tag{A279}$$

We did not explicitly express the dependency on the policy π and the state-action pair (s, a) in the expectation $\mathrm{E}_{s',a'}$. A more precise way would be to write $\mathrm{E}^\pi_{s',a'}\left[. \mid s, a\right]$.
More generally, we have

$$\mathrm{T}^\pi\left[q\right](s,a) \;=\; g(s,a) \;+\; \mathrm{E}_{s',a'}\left[q(s',a')\right] \;. \tag{A280}$$

In the following we show properties for this general formulation.

A7.1.1 Monotonically Increasing and Continuous

We assume the general formulation Eq. (A280) of the Bellman operator. Proposition 2.1 on pages 22-23 in Bertsekas and Tsitsiklis, 1996, [14] shows that a fixed point q^π of the Bellman operator exists and that for every q:

$$q^\pi \;=\; \mathrm{T}^\pi\left[q^\pi\right] \tag{A281}$$

$$q^\pi \;=\; \lim_{k\to\infty} \left(\mathrm{T}^\pi\right)^k q \;. \tag{A282}$$

The fixed point equation

$$q^\pi \;=\; \mathrm{T}^\pi\left[q^\pi\right] \tag{A283}$$

is called *Bellman equation* or *Poisson equation*. For the Poisson equation see Equation 33 to Equation 37 for the undiscounted case and Equation 34 and Equation 43 for the discounted case in Alexander Veretennikov, 2016, [137]. This form of the Poisson equation describes the Dirichlet boundary value problem. The Poisson equation is

$$q^\pi(s,a) \;+\; \bar{g} \;=\; g(s,a) \;+\; \mathrm{E}_{s',a'}\left[q(s',a') \mid s,a\right] \;, \tag{A284}$$

where $\bar{g}$ is the long term average reward or the expected value of the reward for the stationary distribution:

$$\bar{g} \;=\; \lim_{T\to\infty} \frac{1}{T+1} \sum_{t=0}^{T} g(s_t, a_t) \;. \tag{A285}$$

We assume $\bar{g} = 0$ since after some time the agent does no longer receive reward in MDPs with finite time horizon or in MDPs with absorbing states that have zero reward.
T^π is *monotonically increasing* in its arguments [14]. For q_1 and q_2 with the component-wise condition $q_1 \geqslant q_2$, we have

$$\begin{aligned} & \mathrm{T}^\pi\left[q_1\right](s,a) \;-\; \mathrm{T}^\pi\left[q_2\right](s,a) \\ & = \left(g(s,a) \;+\; \mathrm{E}_{s',a'}\left[q_1(s',a')\right]\right) \;-\; \left(g(s,a) \;+\; \mathrm{E}_{s',a'}\left[q_2(s',a')\right]\right) \\ & = \mathrm{E}_{s',a'}\left[q_1(s',a') \;-\; q_2(s',a')\right] \;\geqslant\; 0\;, \end{aligned} \tag{A286}$$

where "$\geqslant$" is component-wise. The last inequality follows from the component-wise condition $q_1 \geqslant q_2$.
We define the norm $\|.\|_\infty$, which gives the maximal difference of the Q-values:

$$\|q_1 \;-\; q_2\|_\infty \;=\; \max_{s,a} |q_1(s,a) \;-\; q_2(s,a)|\;. \tag{A287}$$

T is a *non-expansion mapping* for q_1 and q_2:

$$\begin{aligned} & \|\mathrm{T}^\pi\left[q_1\right] \;-\; \mathrm{T}^\pi\left[q_2\right]\|_\infty \;=\; \max_{s,a} |\mathrm{T}[q_1](s,a) \;-\; \mathrm{T}[q_2](s,a)| \\ & = \max_{s,a} \Bigg| \Bigg[g(s,a) \;+\; \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\, q_1(s',a') \Bigg] \;- \\ & \quad \Bigg[g(s,a) \;+\; \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\, q_2(s',a') \Bigg] \Bigg| \\ & = \max_{s,a} \left| \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s') \;\left(q_1(s',a') \;-\; q_2(s',a')\right) \right| \\ & \leqslant \max_{s,a} \sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s') \;|q_1(s',a') \;-\; q_2(s',a')| \\ & \leqslant \max_{s',a'} |q_1(s',a') \;-\; q_2(s',a')| \;=\; \|q_1 \;-\; q_2\|_\infty\;. \end{aligned} \tag{A288}$$

The first inequality is valid since the absolute value is moved into the sum. The second inequality is valid since the expectation depending on (s,a) is replaced by a maximum that does not depend on (s,a). Consequently, the operator T^π is continuous.

A7.1.2 Contraction for Undiscounted Finite Horizon

For time-aware states, we can define another norm with $0 < \eta < 1$ which allows for a contraction mapping:

$$\|q_1 \;-\; q_2\|_{\infty,t} \;=\; \max_{t=0}^{T} \eta^{T-t+1} \max_{s_t,a} |q_1(s_t,a) \;-\; q_2(s_t,a)|\;. \tag{A289}$$

T^π is a *contraction mapping* for q_1 and q_2 [14]:

$$\|\mathrm{T}^\pi\,[q_1] \;-\; \mathrm{T}^\pi\,[q_2]\,\|_{\infty,t} \;=\; \max_{t=0}^{T}\; \eta^{T-t+1} \max_{s_t,a} |\mathrm{T}[q_1](s_t,a) \;-\; \mathrm{T}[q_2](s_t,a)| \tag{A290}$$

$$= \max_{t=0}^{T}\; \eta^{T-t+1} \max_{s_t,a} \left| \left[g(s_t,a) \;+\; \sum_{s_{t+1}} p(s_{t+1} \mid s_t,a) \sum_{a'} \pi(a' \mid s')\; q_1(s_{t+1},a') \right] - \right.$$

$$\left. \left[g(s_t,a) \;+\; \sum_{s_{t+1}} p(s_{t+1} \mid s_t,a) \sum_{a'} \pi(a' \mid s')\; q_2(s_{t+1},a') \right] \right|$$

$$= \max_{t=0}^{T}\; \eta^{T-t+1} \max_{s_t,a} \left| \sum_{s_{t+1}} p(s_{t+1} \mid s_t,a) \sum_{a'} \pi(a' \mid s')\; [q_1(s_{t+1},a') \;-\; q_2(s_{t+1},a')] \right|$$

$$\leqslant \max_{t=0}^{T}\; \eta^{T-t+1} \max_{s_t,a} \sum_{s_{t+1}} p(s_{t+1} \mid s_t,a) \sum_{a'} \pi(a' \mid s')\; |q_1(s_{t+1},a') \;-\; q_2(s_{t+1},a')|$$

$$\leqslant \max_{t=0}^{T}\; \eta^{T-t+1} \max_{s_{t+1},a'} |q_1(s_{t+1},a') \;-\; q_2(s_{t+1},a')|$$

$$\leqslant \max_{t=0}^{T}\; \eta\, \eta^{T-(t+1)+1} \max_{s_{t+1},a'} |q_1(s_{t+1},a') \;-\; q_2(s_{t+1},a')|$$

$$= \eta \max_{t=1}^{T+1}\; \eta^{T-t+1} \max_{s_t,a'} |q_1(s_t,a') \;-\; q_2(s_t,a')|$$

$$= \eta \max_{t=0}^{T}\; \eta^{T-t+1} \max_{s_t,a'} |q_1(s_t,a') \;-\; q_2(s_t,a')|$$

$$= \eta\, \|q_1 \;-\; q_2\|_{\infty,t}\;.$$

The equality in the last but one line stems from the fact that all Q-values at $t = T+1$ are zero and that all Q-values at $t = 1$ have the same constant value.
Furthermore, all q values are equal to zero for additionally introduced states at $t = T+1$ since for $t > T+1$ all rewards are zero. We have

$$q^\pi \;=\; \mathrm{T}^T\,[q]\;, \tag{A291}$$

which is correct for additionally introduced states at time $t = T+1$ since they are zero. Then, in the next iteration Q-values of states at time $t = T$ are correct. After iteration i, Q-values of states at time $t = T - i + 1$ are correct. This iteration is called the "backward induction algorithm" [95, 96]. If we perform this iteration for a policy π instead of the optimal policy, then this procedure is called "policy evaluation algorithm" [95, 96].

A7.1.3 Contraction for Undiscounted Infinite Horizon With Absorbing States

A stationary policy is *proper* if there exists an integer n such that from any initial state x the probability of achieving the terminal state after n steps is strictly positive.
If all terminal states are absorbing and cost/reward free and if all stationary policies are proper the Bellman operator is a contraction mapping with respect to a weighted sup-norm.
The fact that the Bellman operator is a contraction mapping with respect to a weighted sup-norm has been proved in Tseng, 1990, in Lemma 3 with equation (13) and text thereafter [132]. Also Proposition 1 in Bertsekas and Tsitsiklis, 1991, [13], Theorems 3 and 4(b) & 4(c) in Tsitsiklis, 1994, [133], and Proposition 2.2 on pages 23-24 in Bertsekas and Tsitsiklis, 1996, [14] have proved the same fact.

A7.1.4 Fixed Point of Contraction is Continuous wrt Parameters

The mean q^π and variance V^π are continuous with respect to π, that is $\pi(a' \mid s')$, with respect to the reward distribution $p(r \mid s,a)$ and with respect to the transition probabilities $p(s' \mid s,a)$.
A complete metric space or a Cauchy space is a space where every Cauchy sequence of points has a limit in the space, that is, every Cauchy sequence converges in the space. The Euclidean space $\mathbb{R}^n$ with the usual distance metric is complete. Lemma 2.5 in Jachymski, 1996, is [62]:

Theorem A11 (Jachymski: complete metric space)**.** *Let (X,d) be a complete metric space, and let (P,d_P) be a metric space. Let $F : P \times X \to X$ be continuous in the first variable and contractive*